(1) $G(s) = \frac{80(s+12)(s+10)}{s^2(s+100)(s+1000)} = \frac{80(12)(10)\left(\frac{s}{12}+1\right)\left(\frac{s}{10}+1\right)}{(100)(1K)\,s^2\left(\frac{s}{100}+1\right)\left(\frac{s}{1K}+1\right)}$

$G(jw) = \frac{0.096\left(\frac{jw}{12}+1\right)\left(\frac{jw}{10}+1\right)}{(jw)^2\left(\frac{jw}{100}+1\right)\left(\frac{jw}{1000}+1\right)}$

a) $K = 0.096$

b) 1) $20\log(0.096) = -20.35\,dB$

K shifts mag plot down by $-20.35\,dB$

2) Because of jw^2, the slope at the beginning is $-40\,dB/dec$

3) Z_1 causes the slope to rise by $20\,dB/dec$ at $w = 12$

4) Z_2 causes the slope to rise by $20\,dB/dec$ at $w = 10$

5) $p1$ cause the slope to fall by $20\,dB/dec$ at $w = 100$.

6) $p2$ causes the slope to fall by $20\,dB/dec$ at $w = 1000$

c) 1) k has no effect

2) jw^2 has the plot start at -180

3) z_1 has a +90° shift from 1 to 100

4) z_2 has a +90° shift from 1.2 to 120

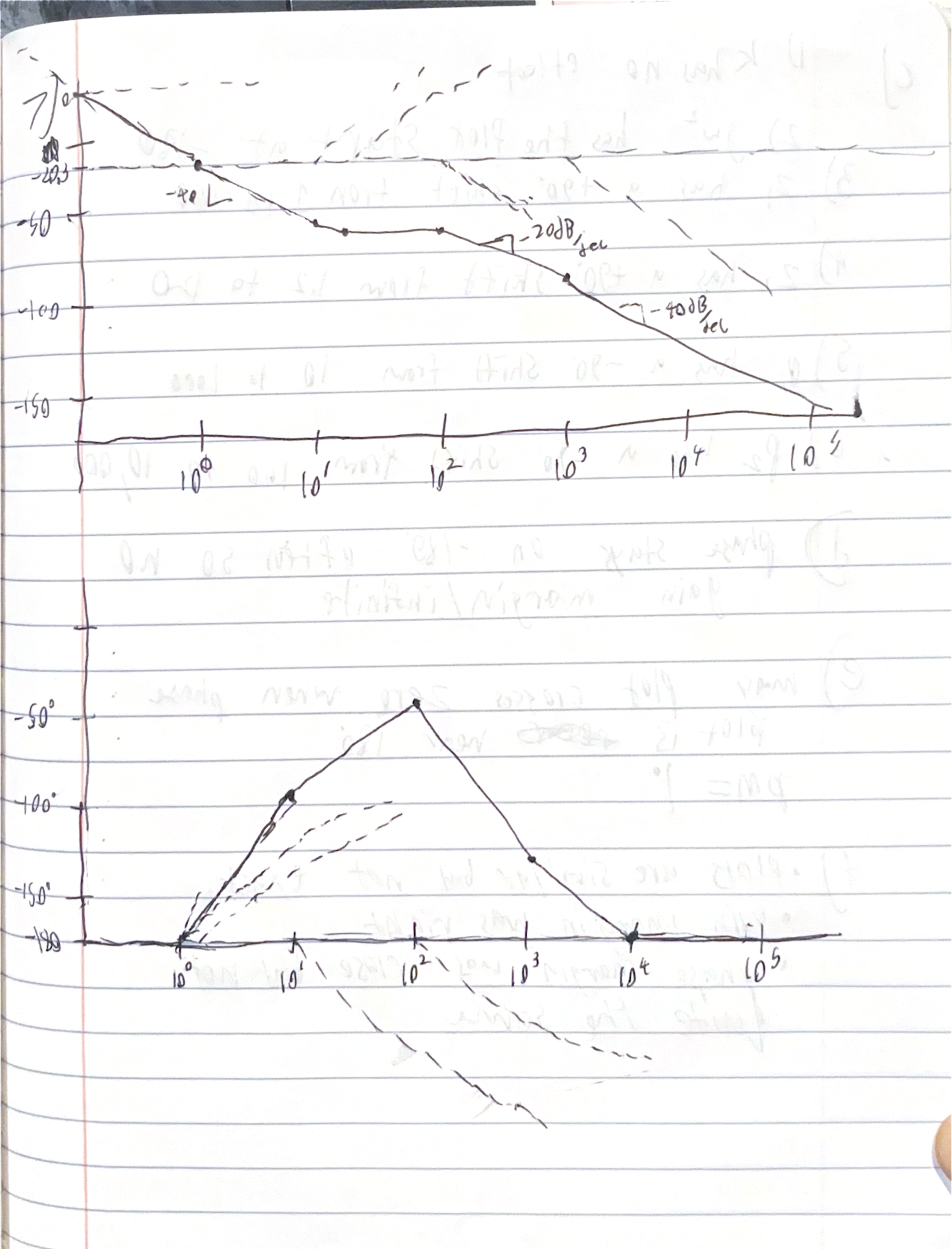
5) p_1 has a -90° shift from 10 to 1000

6) p_2 has a -90° shift from 100 to 10,000

d) phase stays on -180° often so no gain margin/infinite

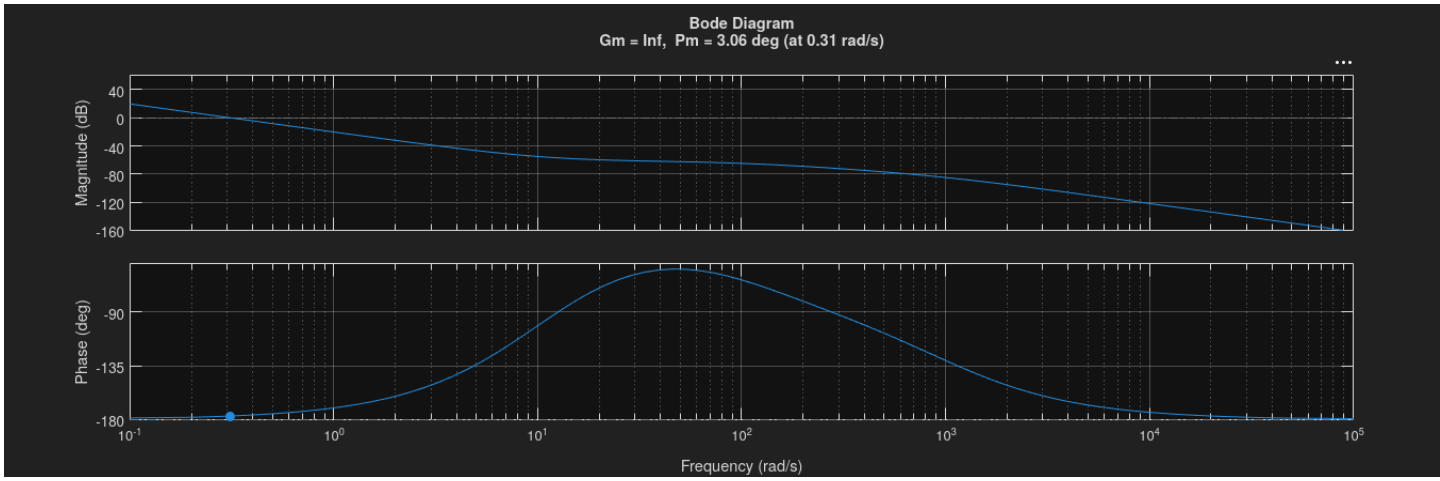
e) mag plot crosses zero when phase plot is ~~zero~~ near 180°
pm = 1°

f) • plots are similar but not exact.
• gain margin was right
• phase margin was close but not quite the same


0
-20
-40
-40dB/dec
-20dB/dec
-40dB/dec
-100
-150
10^0
10^1
10^2
10^3
10^4
10^5
-50°
-100°
-150°
-180
10^0
10^1
10^2
10^3
10^4
10^5

7.

f)


Bode Diagram
Gm = Inf, Pm = 3.06 deg (at 0.31 rad/s)
Magnitude (dB)
40
0
-40
-80
-120
-160
Phase (deg)
-90
-135
-180
10^-1
10^0
10^1
10^2
10^3
10^4
10^5
Frequency (rad/s)